

RMOKV– 2024

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Find all positive integer x, y such that $202x + 4x^2 = y^2$.
2. Show that there do not exist non-zero real numbers a, b, c such that the following statements hold simultaneously:
 - (a) The equation $ax^2 + bx + c = 0$ has two distinct roots x_1, x_2
 - (b) The equation $bx^2 + cx + a = 0$ has two distinct roots x_2, x_3
 - (c) The equation $cx^2 + ax + b = 0$ has two distinct roots x_3, x_1
 (Note that x_1, x_2, x_3 may be real or complex numbers.)
3. Let ABC be an equilateral triangle. Suppose D is a point on BC such that $BD : DC = 1:3$. Let the perpendicular bisector of AD intersect AB, AC , at E, F respectively. Prove that $49[BDE]=25[CDF]$, where $[XYZ]$ denotes the area of the triangle XYZ .
4. Let $n>1$ be an integer. Call a rearrangement $a_1, a_2, a_3, \dots, a_n$ of $1, 2, 3, \dots, n$ nice if for every $k = 2, 3, 4, \dots, n$ we have $a_1 + a_2 + a_3 + \dots + a_k$ is not divisible by k . Determine which positive integer $n>1$ have a nice rearrangement.
5. Let ABC be a triangle with $\angle ABC = 20^\circ, \angle ACB = 40^\circ$. Let D be a point on BC such that $\angle BAD = \angle DAC$. Let the incircle of triangle ABC touch BC at E . Prove that $BD = 2.CE$.
6. Let X be a set of 11 integers. Prove that one can find a nonempty subset $\{a_1, a_2, a_3, \dots, a_k\}$ of X such that 3 divides k and 9 divides the sum $\sum_{i=1}^k 4^i a_i$

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